

Assignment 1. DTU-Autumn 2025

02429 Analysis of correlated data: Linear mixed models

September 17, 2026

To be uploaded to dtulearn by **Mondayay the 28th by 9:00**

To be solved in groups of 2 to 3 students, groups of size 1 are NOT accepted.

The report is roughly composed as follows:

- i) A **maximum of 12 pages** for the main text.
- ii) Include as many appendices as necessary for additional graphs and tables of interest.
- iii) Include one appendix with the output for `ranova(final_model)`, `drop1(final_model, test="F")`, `summary(final_model)`. These will allow us to verify the results. If the output of `summary(final_model)` is too long, include an abbreviated version (fewer rows).
- iv) An appendix with the full R code at the end of the report.

Remember: use no more than four significant digits and fewer if possible. Use a good font size for numbers and labels in the figures. All figures and tables must be referred to in the text, should be numbered and have an informative legend. Balance the length of the different sections (exploratory analysis, modelling, ..., conclusions) in the main text. When writing your solutions consider that the readers are the class members.

The solution of each exercise can roughly follow the following steps.

I. Exploratory analysis.

- Do a graphical and numerical exploratory analysis. What do you see?

II. Modelling.

- Start with a model and build up or down a potential final model.
- Present diagnostic plots/statistics of the final model
- Give a mathematical expression of the starting and final model.

III Results.

- Give the numerical results that answer the questions
- When presenting parameter estimates remember to include confidence intervals.
- Present any additional results you may consider relevant or informative.

Exercise I. Experimental data on heritability. A components of variance model.

In research on artificial insemination of cows, a series of semen samples from a bull are tested for their ability to produce conceptions. The following data show the percentage of conceptions obtained from samples of six bulls.

Table 1: Percentage of conceptions to Services for successive samples, y_{ij}

Bull, i	y_{ij}	n_i	$\sum_j^{n_j} y_{ij}$	$\bar{y}_i$	s_i^2
1	46, 31, 37, 62, 30	5	206	41.20	175.7
2	70, 59	2	129	64.50	60.5
3	52, 44, 57, 40, 67, 64, 70	7	394	56.29	132.9
4	47, 21, 70, 46, 14	5	198	39.60	505.3
5	42, 64, 50, 69, 77, 81, 87	7	470	67.14	270.5
6	35, 68, 59, 38, 57, 76, 57, 29, 60	9	479	53.22	249.4
Total		35	1876	321.95	

Snedecor & Cochran, Statistical methods, p. 246

Objective: Estimate the percentage of conceptions with a confidence interval. Report the variability in the percentages due to bull-to-bull variability and within-bull variability (components of variance). And the heritability parameter.

The six bulls are considered to be a *random sample* from a population of bulls. The percentages of conceptions for the same bull are assumed to be correlated, but from one bull to another are assumed to be independent.

Exercise II. Experimental data on surgery.

The data shown in the Table below have been previously analyzed by P. McCullagh. The data were obtained in an experiment with the purpose to investigate the effect of hyperbaric O2 treatment on the healing of surgical wounds in rats. Thirty rats underwent surgery, during which an incision was made along the entire length of the back. This was immediately sewn up with surgical staples, and the rats were returned to their cages.

The treatment group of fifteen rats was subjected to hyperbaric O2 treatment, i.e., a 100% oxygen environment at two atmospheres of pressure, for 90 minutes per day following surgery. The control group also received a similar treatment for 90 minutes daily, but at standard oxygen concentration and normal atmospheric pressure. After 24 days, strips of skin were taken from five sites labelled AE (front to back) on each rat. The strips were put on a tensiometer, stretched to the breaking point, and the energy required to break the strip was recorded. Six rats were excluded from the analysis as they did not fulfil the experiment protocol. Some measurements could not be made, they are indicated by – in the table. Rats 114 received the hyperbaric treatment, rats 1524 were the controls. The rats varied in size and pre- and post-surgery weight, but these measurements were not recorded.

The main objective is to determine whether or not hyperbaric O2 treatment has an effect on the healing of surgical wounds, and if so, whether the effect depends on the position of the wound on the back. The data are in datafile `read.csv("Data/RatsMcCullagh.csv")` or

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load("Data/RatsMcCullagh.RData").
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Exercise:

- i) Fit a linear mixed model to investigate whether site and treatment have a statistically significant effect on the logarithm of the response variable.
- ii) Suppose that the researcher suspects that the site next to the tail (site "5"), is the weakest section for the skin to break. Can you justify it using a fitted model?
- iii) There are 104 measurements (observational units) belonging to 24 rats (24 experimental units). What happens if you ignore the fact that some measurements are made on the same rat and use the 104 experimental observations as if they were 104 independent observations? Does site or treatment show a statistically significant effect on the logarithm of the response?

Table 2: Strength of skin strips 24 days after surgery
Rat Site on the back: shoulder to tail

Rat	A	B	C	D	E	Mean
1	3.8300	7.3788	44.353	19.555	–	18.779
2	27.861	29.974	15.470	23.455	–	24.190
3	56.996	60.960	20.306	–	28.123	41.596
4	–	38.043	68.080	42.425	30.335	44.721
5	16.276	–	59.033	73.891	–	49.733
6	38.267	33.702	35.558	44.598	32.678	39.961
7	9.0384	11.259	27.121	31.984	–	19.851
8	16.728	27.590	13.238	12.139	6.3865	15.216
9	11.866	27.983	26.226	15.594	19.225	20.179
10	23.352	34.790	27.556	35.883	22.848	28.888
11	16.444	31.928	21.495	15.590	7.0750	18.506
12	23.342	46.313	33.810	15.686	–	29.788
13	15.267	14.452	10.635	22.156	6.8062	13.863
14	21.732	20.746	12.293	17.295	10.301	16.473
15	82.508	13.645	49.187	–	53.432	49.693
16	–	45.919	63.090	68.137	36.500	53.412
17	80.147	29.943	71.928	–	46.609	57.157
18	31.938	–	36.211	49.815	44.468	40.608
19	15.453	31.384	27.127	27.961	9.9035	22.366
20	21.183	27.429	20.058	–	–	22.890
21	20.445	12.532	15.661	28.694	–	19.333
22	16.928	59.579	29.407	18.626	8.8352	26.675
23	35.631	21.613	23.155	42.379	16.203	27.796
24	20.523	24.621	16.292	–	18.680	20.029
Mean	27.534	29.627	31.970	31.888	23.436	29.126

Rats 1–14 are hyperbaric O₂-treated; rats 15–24 are the controls. Data analyzed by P. McCullagh.

Exercise III. Experimental data on saws.

This example is taken from Bliss and reanalyzed by P. McCullagh. It consists of an experiment designed to compare the performance of three different brands of chain saw.

The woodcutting efficiencies of three brands of saw were compared in a fractional factorial design using six cutting teams, three species of softwood (spruce, pine and larch) both with bark and without bark. The response variable is the time in minutes taken to complete a designated cutting task.

The fractional factorial is embedded in a 6×6 Latin square whose columns correspond to six teams of workmen covering the range from experienced woodcutters to seasonal labourers. The Roman letters correspond to six distinct saws, where A, D are duplicates of brand 1, B, E are duplicates of brand 2, and C, F are duplicates of brand 3. The table below shows the design and the response values.

The main purpose of the study is to compare the relative efficiencies of the three brands of saw. There are 12 observations for each brand, and the sample averages are 8.78, 8.55 and 7.42 in minutes, so brand 3 appears to be the most efficient. The main statistical challenge is to provide a reasonable assessment of the standard error for brand contrasts (differences).

In addition to brand effects, we can also investigate the effect of de-barking. The sample averages with and without bark are 8.80 and 7.70 minutes, or 2.13 and 1.97 on the log scale, so de-barking appears to reduce the cutting time by about 12–15%. Is the reduction approximately the same for each saw brand?

In addition to brand and debarking effects, we can also investigate differences between the three species. There are 12 observations for each species, and the average cutting times for spruce, pine and larch are 8.03, 6.42 and 10.30 minutes respectively, or 2.04, 1.82 and 2.29 on the log scale. Larch, one of the few deciduous conifers, is evidently substantially harder or tougher than the other two. Regardless of whether we use the log scale or the time scale for averages, how do we calculate a standard error for species contrasts?

The data are in datafile `read.csv("data/SawBliss.csv")` or in `load("data/SawBliss.RData")`.

Table 3: Time in minutes taken by six teams to complete a woodcutting task using one of six available saws A–F. From Bliss (1970) with one correction in row 1, col. 6. (P. McCullagh)

Species	Bark	1	2	3	4	5	6
Spruce	No	6.4 F	10.9 E	9.8 D	7.5 B	4.6 A	4.9 C
Pine	No	6.8 B	6.2 C	7.9 E	6.0 A	4.0 D	4.2 F
Larch	No	12.7 E	13.4 A	12.5 B	7.3 C	6.1 F	7.4 D
Spruce	Yes	8.8 C	10.2 D	12.5 A	8.6 F	6.1 E	6.0 B
Pine	Yes	7.4 D	10.0 B	8.3 F	6.4 E	4.3 C	5.6 A
Larch	Yes	13.1 A	12.0 F	12.0 C	11.3 D	6.1 B	9.7 E

Exercise:

Fit a linear mixed model to answer the questions of interest. The teams are meant to be considered as a sample of teams, and so are the saws.

- i) Compare the efficiencies of the three saw brands.
- ii) Estimate the differences in cutting times according to the debarking. Are they statistically different?
- iii) Estimate the differences in cutting times due to species.